

D1 (IO, 3 W)

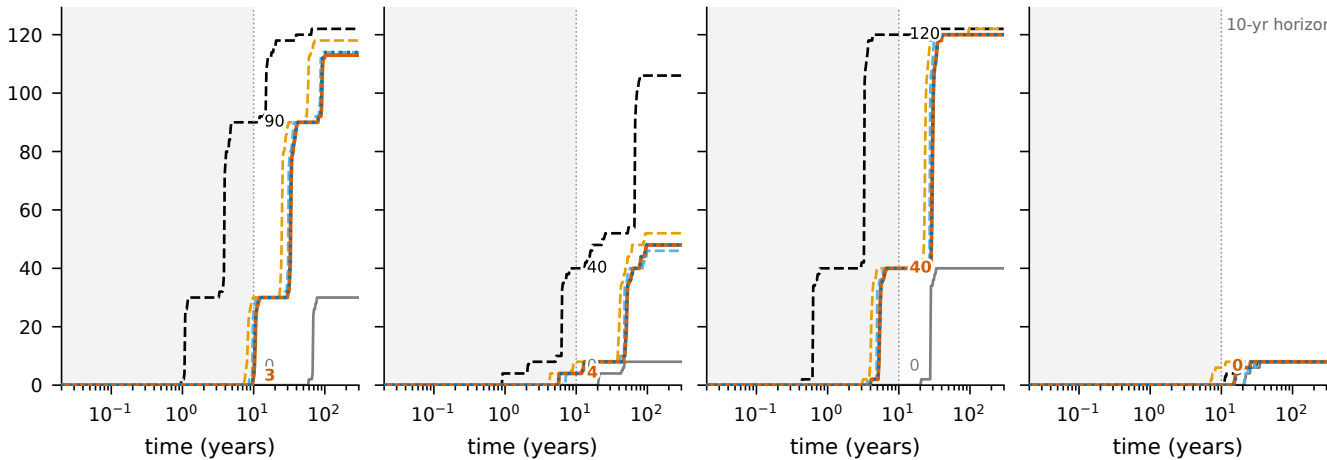
D2 (logic, 12 W)

D3 (SRAM, 6 W)

D4 (logic, 25 W)

10-yr horizon

rails nucleated



— die-level sign-off (standalone thermal)
 - - - rule sign-off (uniform 105 °C, const. σ_T)

- - - uniform die temperature
 - - - σ_T frozen at 2-D temperature

- - - no wire self-heating
 - - - no thermomigration

— full 3-D physics (FDM truth)
 ... StackEM (SKN + JCX)